Formalizing the Genetic Code: Defining Metrics for Geometric and Topological Sequence Analysis

I. Introduction: The Biosemiotics of DNA and the Requirement for Formal Metrics

The analysis of Deoxyribonucleic Acid (DNA) structure through the lens of formal language theory and biosemiotics has provided a powerful methodological exchange between biology, linguistics, and computer science, fostering the field of Biolinguistics. Since the discovery of the DNA double helix in 1953, the idea of the genetic code as a linguistic structure, sharing fundamental units, operations, and structures with natural language, has driven research into biological sequence interpretation. To transition this conceptual analogy into rigorous computational models suitable for geometric analysis—specifically, projecting the sequence space onto a hyperspace for topological feature capture—formal, quantifiable definitions of the elemental, sequential, and hierarchical relationships are required.

This report establishes the necessary foundational metrics for encoding the genetic information: the Signary Map, defining the vector representation of the elementary signs (codons); the Positional Metric, governing the sequential relationship and directionality; and the Compositional Metric, defining the syntactic rules for grouping signs into functional units. The objective is to provide a consistent, mathematically implementable framework that translates the quaternary alphabet of DNA into a binary vector space ($\mathbb{B}^6$) suitable for 2^6 hypercube analysis.

The Scope and Necessity for Geometric Vectorization

The genetic code utilizes 64 distinct codons (4^3), requiring 6 bits to assign a unique identifier to each triplet ($2^6 = 64$). Mapping these 64 signs into a 6-bit vector provides the coordinates for a 6-dimensional Hamming space, allowing algebraic and topological techniques—such as the Category-based Topological Sequence Analysis (CTSA)—to capture structural information and hierarchical organization across multiple scales.

A critical consideration, however, is the disparity between arbitrary geometric distance and inherent biological distance. While an arbitrary binary map provides necessary coordinates, studies have demonstrated that such mappings often fail to preserve the intrinsic organization and symmetries of the genetic code. The genetic code is robust and highly optimized to mitigate the effects of point mutations. For instance, a biological 1-distance (a transition or transversion mutation between two codons, such as UUU to UUC) should ideally correspond to a 1-distance in the binary Hamming space (only one bit change). However, arbitrary numerical schemes frequently result in a 2-distance. Therefore, while a consistent numerical index is provided as requested for immediate geometric computation, researchers utilizing this framework must recognize that the geometric relationships in the raw 2^6 space may distort biologically relevant topological relationships unless the metric is weighted or explicitly designed to reflect physicochemical properties.

II. The Signary Map: Constructing the 6-Bit Codon Vectorization

The Signary Map formalizes the translation of the DNA quaternary alphabet (\mathcal{A}=\{A, C, G, T\}) into a fixed-length binary word. For 64 codons, a 6-bit vector is required, where the vector C = (b_1 b_2 b_3 b_4 b_5 b_6) serves as the coordinate of the sign in the \mathbb{B}^6 hypercube.

Base-Level Encoding Rule

To construct the lexicographical map such that the codon sequence proceeds in a consistent numerical order, the four DNA bases (Adenine, Cytosine, Guanine, Thymine) must first be assigned a 2-bit value. To satisfy the requirement that the lowest index codon (AAA) maps to the lowest binary vector (000000), Adenine (A) must be assigned the lowest binary value, and the sequence of assignment must follow a standard lexicographical order corresponding to increasing binary value:

1. A (Adenine) \to 00
2. C (Cytosine) \to 01
3. G (Guanine) \to 10
4. T (Thymine) \to 11

Codon Construction and Lexicographical Mapping

A codon C = N_1 N_2 N_3 is constructed by concatenating the 2-bit representations of the three constituent nucleotides: C = (N_1^{2-bit} N_2^{2-bit} N_3^{2-bit}).
For example:

- AAA \to (00)(00)(00) \to 000000 (Decimal Index 0)
- TGT \to (11)(10)(11) \to 111011 (Decimal Index 59)

The complete 6-bit lexicographical Signary Map for all 64 DNA codons, indexed from 0 to 63, is provided below. This map forms the fundamental input for the geometric analysis.
Table: The Lexicographical 6-Bit Signary Map of the 64 Codons

Codon (5' → 3')	**Base Encoding (\mathbf{N_1} \mathbf{N_2} \mathbf{N_3})**	**Decimal Index**	**6-Bit Binary Vector**	**Amino Acid**
AAA	00 00 00	0	000000	Lysine (K)
AAC	00 00 01	1	000001	Asparagine (N)
AAG	00 00 10	2	000010	Lysine (K)
AAT	00 00 11	3	000011	Asparagine (N)
ACA	00 01 00	4	000100	Threonine (T)
ACC	00 01 01	5	000101	Threonine (T)
ACG	00 01 10	6	000110	Threonine (T)
ACT	00 01 11	7	000111	Threonine (T)
AGA	00 10 00	8	001000	Arginine (R)
AGC	00 10 01	9	001001	Serine (S)
AGG	00 10 10	10	001010	Arginine (R)

Codon (5' → 3')	**Base Encoding (\mathbf{N_1} \mathbf{N_2} \mathbf{N_3})**	**Decimal Index**	**6-Bit Binary Vector**	**Amino Acid**
AGT	00 10 11	11	001011	Serine (S)
ATA	00 11 00	12	001100	Isoleucine (I)
ATC	00 11 01	13	001101	Isoleucine (I)
ATG	00 11 10	14	001110	Methionine (M, START)
ATT	00 11 11	15	001111	Isoleucine (I)
CAA	01 00 00	16	010000	Glutamine (Q)
CAC	01 00 01	17	010001	Histidine (H)
CAG	01 00 10	18	010010	Glutamine (Q)
CAT	01 00 11	19	010011	Histidine (H)
CCA	01 01 00	20	010100	Proline (P)
CCC	01 01 01	21	010101	Proline (P)
CCG	01 01 10	22	010110	Proline (P)
CCT	01 01 11	23	010111	Proline (P)
CGA	01 10 00	24	011000	Arginine (R)
CGC	01 10 01	25	011001	Arginine (R)
CGG	01 10 10	26	011010	Arginine (R)
CGT	01 10 11	27	011011	Arginine (R)
CTA	01 11 00	28	011100	Leucine (L)
CTC	01 11 01	29	011101	Leucine (L)
CTG	01 11 10	30	011110	Leucine (L)
CTT	01 11 11	31	011111	Leucine (L)
GAA	10 00 00	32	100000	Glutamate (E)
GAC	10 00 01	33	100001	Aspartate (D)
GAG	10 00 10	34	100010	Glutamate (E)
GAT	10 00 11	35	100011	Aspartate (D)
GCA	10 01 00	36	100100	Alanine (A)
GCC	10 01 01	37	100101	Alanine (A)
GCG	10 01 10	38	100110	Alanine (A)
GCT	10 01 11	39	100111	Alanine (A)
GGA	10 10 00	40	101000	Glycine (G)
GGC	10 10 01	41	101001	Glycine (G)
GGG	10 10 10	42	101010	Glycine (G)
GGT	10 10 11	43	101011	Glycine (G)
GTA	10 11 00	44	101100	Valine (V)
GTC	10 11 01	45	101101	Valine (V)
GTG	10 11 10	46	101110	Valine (V)
GTT	10 11 11	47	101111	Valine (V)
TAA	11 00 00	48	110000	STOP (Ter)
TAC	11 00 01	49	110001	Tyrosine (Y)
TAG	11 00 10	50	110010	STOP (Ter)

Codon (5' → 3')	Base Encoding (\mathbf{N_1} \mathbf{N_2} \mathbf{N_3})	Decimal Index	6-Bit Binary Vector	Amino Acid
TAT	11 00 11	51	110011	Tyrosine (Y)
TCA	11 01 00	52	110100	Serine (S)
TCC	11 01 01	53	110101	Serine (S)
TCG	11 01 10	54	110110	Serine (S)
TCT	11 01 11	55	110111	Serine (S)
TGA	11 10 00	56	111000	STOP (Ter)
TGC	11 10 01	57	111001	Cysteine (C)
TGG	11 10 10	58	111010	Tryptophan (W)
TGT	11 10 11	59	111011	Cysteine (C)
TTA	11 11 00	60	111100	Leucine (L)
TTC	11 11 01	61	111101	Phenylalanine (F)
TTG	11 11 10	62	111110	Leucine (L)
TTT	11 11 11	63	111111	Phenylalanine (F)

Encoding Biological Constraints into the Signary Map

For more advanced geometric applications, particularly those seeking to identify meaningful clusters corresponding to synonymous codons or physicochemical similarity, an encoding that reflects the structure of the genetic code is often preferred over simple lexicographical indexing. Certain schemes specifically allocate bits to biologically relevant features.
For instance, one robust strategy involves dedicating the first four bits of the 6-bit vector (b_1 b_2 b_3 b_4) to encode the physicochemical properties of the resultant amino acid or the nucleotide structure of the first two codon positions (N_1 N_2). The final two bits (b_5 b_6) are reserved exclusively for the identity of the third base (N_3). This separation is purposeful: it incorporates the biological constraint of the wobble hypothesis, where changes in N_3 often result in synonymous codons. By isolating N_3 to the lowest-order bits, the geometric space naturally clusters codons based on the primary determinants of amino acid identity (N_1 N_2) and allows the geometric analysis to inherently account for the small conformational adjustments affecting anticodon pairing. Implementing such a scheme ensures that the geometric distance metric is correlated with the functional distance, thereby enriching the interpretability of the resulting topological features.

III. The Positional Metric: Sequentiality and Biological Directionality

The Positional Metric defines the linear relationship between signs (codons) in the sequence. In the context of linguistic analogy, this is the concatenation rule. In DNA, this rule is governed by the rigid constraints of biological transcription and translation, determined by directionality and the reading frame.

Defining Sequential Concatenation via the Reading Frame

The fundamental constraint defining the sequential metric is the **reading frame** (RF). An RF is

the specific choice among the possible ways to read a sequence of nucleotides as non-overlapping triplets. These triplets, when relevant to translation, are the codons. The sequential relationship is fixed by the directionality: the sequence is read from the 5'-phosphoryl end to the 3'-hydroxyl end.

For any single strand of DNA, there are three possible reading frames, determined by which of the first three nucleotides initiates the triplet sequence. In double-stranded DNA, this means there are six potential reading frames in total—three on the sense strand (5' \to 3') and three on the complementary (anti-parallel) strand (5' \to 3' along that strand, corresponding to the 3' \to 5' direction on the first strand). The biologically relevant Positional Metric is defined only by the Open Reading Frame (ORF), which begins with a START codon (typically ATG) and proceeds sequentially in the 5' \to 3' direction until a STOP codon (TAA, TAG, TGA) is encountered.

The linear Positional Metric D_{Lin} between two codons C_i and C_j within a confirmed ORF is simply the absolute difference in their ordered indices:

Elevating the Metric: Integrating Positional Information Theory

However, simple linear distance only accounts for syntactic order. For a comprehensive geometric analysis of biological function, the metric must quantify the functional significance of position within the sequence environment. This necessity moves the definition of the Positional Metric beyond simple concatenation toward the framework of **Positional Information (PI)** theory, initially developed in developmental biology.

Positional Information theory holds that the functional fate (e.g., cell specification or gene expression, g) is stochastically encoded by the molecular cues (local concentrations, sequence motifs) found at a specific position (x). This concept provides a rigorous, information-theoretic methodology for quantifying the significance of position.

For geometric analysis, the positional metric (D_{Pos}) must integrate the linear separation (\delta_{i, j}) with the informational capacity of that position, specifically incorporating the uncertainty inherent in the mapping from sequence position to function. This uncertainty is quantified using Shannon entropy, S.

In this formulation, f(\delta_{i, j}) is a function of the linear distance (e.g., |i - j|), and S_c represents the contextual Shannon entropy—the measure of remaining uncertainty regarding the functional assignment of positions i and j given their local environment. If a position has high uncertainty (S_c is high, such as a position within non-coding junk DNA or a highly ambiguous regulatory region), the \exp(-S_c) term reduces the weight of that distance measurement, indicating low certainty about the functional connection between C_i and C_j. Conversely, positions with highly deterministic functional roles (low S_c, e.g., an active site codon in a highly conserved exon) will contribute greater functional weight to the overall distance metric, regardless of whether the linear distance is large or small. Therefore, the refined Positional Metric ensures that geometric analysis is intrinsically weighted by the informational reliability of the sequence location, transforming the coordinate system from purely physical space to information space.

IV. The Compositional Metric: Syntactic Hierarchy and Functional Grouping

The Compositional Metric defines the syntactic relationship among codons, specifying the rules by which codons are grouped into higher-level functional units, such as regulatory motifs, exons,

genes, or operons. This metric defines the "sentences" and "paragraphs" of the genomic language.

Syntactic Complexity and Context Dependence

Unlike simple linear languages, the interpretation of DNA sequences is profoundly context-dependent. The functional meaning (semantics) of a sequence element often relies entirely on its surrounding compositional environment and location. For example, the recognition of a promoter sequence or a splice site is not an intrinsic property of the short motif itself, but rather a function of the context provided by adjacent sequence elements (e.g., upstream and downstream markers, protein binding sites). This characteristic fundamentally differentiates the syntax of DNA from simpler linguistic models.

Consequently, standard Context-Free Grammars (CFGs), which are widely used in computational linguistics, are inadequate for modeling genomic structure because they cannot capture the necessity for context in assigning function. The syntax of DNA, especially in eukaryotes, exhibits a nested, hierarchical structure, referred to as "boundaries within boundaries," which defines complex organizational levels, from individual functional domains within proteins to large "swarm agents" like populations and communities.

Formal Definition using Context-Sensitive Grammars (CSG)

The Compositional Metric must therefore be formally defined by the production rules of a **Context-Sensitive Grammar (CSG)**, which possesses the necessary expressive power to model these structural and positional variabilities.

A CSG rule takes the form $\alpha A \beta \to \alpha \gamma \beta$, where the non-terminal symbol A can only be rewritten as γ if it is preceded by the context α and followed by the context β.

The Compositional Metric establishes the functional boundaries of genomic units based on the satisfaction of these context-sensitive production rules, ensuring that biological "meaning" arises from specific sequence environments.

Functional Grouping and Syntactic Validity

The primary compositional units defined by this metric are:

1. **Genes/Operons:** Defined as structured sequences starting with an initiation sequence (e.g., ATG) and ending with a termination sequence (STOP codons), bounded by regulatory elements like promoters and terminators.
2. **Regulatory Elements and Splice Sites:** These elements require CSG to define their functional grouping. Consider the canonical eukaryotic donor splice site pattern, typically AGGT...CAG. The Compositional Metric dictates that the sequence pattern AGGT only constitutes a functional *donor site* (and thus a grouping boundary) if the CSG rule is satisfied: it must be preceded by an **Exon** context (α) and followed by an **Intron** context (β).

The Compositional Metric M_{Comp} quantifies the likelihood of a codon C_i belonging to a specific functional unit U based on the satisfaction of the CSG rule set $\mathcal{G}_{CSG}$:

This framework implies that the syntactic structure (the Compositional Metric) actively determines the semantic function. For geometric analysis, M_{Comp} serves as a crucial filter, segmenting the linear sequence into functionally coherent groups (e.g., identifying the span of a

gene, or the precise boundaries of a regulatory domain). This filtering ensures that topological analysis focuses only on syntactically valid "sentences," maximizing the predictability, modularity, and biological fidelity of the resulting geometric interpretation. Furthermore, this grammar-based approach provides a foundation for designing synthetic DNA constructs by incorporating linguistic constraints to maximize intended regulatory behavior.

V. Conclusions and Synthesis for Geometric Analysis

The formalization of DNA as a linguistic structure requires the rigorous definition of the Signary, Positional, and Compositional Metrics to enable sophisticated geometric and computational analysis.

The **Signary Map** provides the foundation by establishing a consistent 6-bit binary vector for each of the 64 codons, translating the sequence into coordinates within a 6-dimensional Hamming space. While the lexicographical scheme provided serves as a simple coordinate system, geometric analysis aimed at interpreting biological relationships should utilize an alternative physicochemical encoding that better preserves the optimized symmetries and mutation-robustness of the canonical genetic code.

The **Positional Metric** extends beyond simple linear concatenation along the Open Reading Frame (5' \to 3' direction). By integrating concepts from Positional Information theory and Shannon entropy, the metric ensures that the relative "distance" between two codons in the geometric space is not solely based on physical separation, but is weighted by the information content and functional certainty of their respective locations.

The **Compositional Metric**, defined by Context-Sensitive Grammars (CSG), provides the crucial syntactic overlay, segmenting the sequence into meaningful, hierarchical functional units (e.g., genes, promoters, splice sites). This metric prevents the geometric analysis from treating functionally distinct or syntactically invalid segments (such as non-coding regions or alternative frames) equally.

When integrated, these three metrics define a rigorous methodology for geometric biolinguistics. The 6-bit vectors provide the fundamental data representation, the Positional Metric imposes informational weighting, and the Compositional Metric enforces biologically constrained boundaries. This structured approach directly supports advanced analytical methods, such as topological sequence analysis (CTSA) , enabling the robust capture of the hierarchical organization and structural features of the genetic sequence at multiple functional scales. The ultimate goal is to move the analysis of genomic sequences from linear textual interpretation into the realm of algebraic geometry, where sequence complexity can be quantified and compared through topological features.

Works cited

1. Natural language and the genetic code: from the semiotic analogy to biolinguistics - CORE, https://core.ac.uk/download/pdf/61912551.pdf 2. Topological sequence analysis of genomes: category theory approaches - arXiv, https://arxiv.org/html/2507.08043 3. The Graph, Geometry and Symmetries of the Genetic Code with Hamming Metric - MDPI, https://www.mdpi.com/2073-8994/7/3/1211 4. Some theoretical aspects of reprogramming the standard genetic code - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC8128387/ 5. Effective Variations of related Physicochemical Properties of Nucleotides Leading to Amino Acids for Characterizing Genes and Proteins - bioRxiv,

https://www.biorxiv.org/content/10.1101/332171v2.full.pdf 6. Reading frame - Wikipedia, https://en.wikipedia.org/wiki/Reading_frame 7. DNA and RNA codon tables - Wikipedia, https://en.wikipedia.org/wiki/DNA_and_RNA_codon_tables 8. The many bits of positional information - PMC - PubMed Central, https://pmc.ncbi.nlm.nih.gov/articles/PMC7875500/ 9. Positional information, in bits - PNAS, https://www.pnas.org/doi/10.1073/pnas.1315642110 10. Deciphering the linguistic blueprint of DNA: context-sensitive ... - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC12551216/ 11. Functional Information: Towards Synthesis of Biosemiotics and Cybernetics - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC3285384/